

CHAPTER ONE: SAMPLING AND SAMPLING DISTRIBUTIONS

1.1 Basic Concepts

Population: is the complete collection of individuals, objects or measurements for which inferences are to be made. The population represents the target of an investigation, and the objective of the investigation is to draw conclusions about the population and it should be defined on the basis of the objective of the study by the investigator.

Example 1.1

- All customers of electric supply company.
- All students of JKU
- Population of farms having a certain type of natural fertility.
- Population of households in a certain village.

Sample: A sample from a population is the set of measurements that are actually collected in the course of an investigation. It should be selected using some predefined sampling technique in such a way that they represent the population very well.

Sampling (elementary) unit: - the ultimate unit to be sampled or elements of the population to be sampled.

Example 1.2: If somebody studies economic status of the households, households is the sampling unit. If one studies performance of freshman students in some college, the student is the sampling unit.

Sampling frame: is the list of all elements (sampling units) in a population.

Example 1.3:

- ✓ List of households of a certain city.
- ✓ List of students in the registrar office of the university.

Parameter and Statistic are basic terms in sampling theory. Parameter is a value calculated from the population. For instance population mean, population variance, population proportion is parameters. Statistic is a value calculated from a sample. Sample mean, sample variance, sample proportion, etc are statistics.

Sampling error: A type of error that may arise due to inappropriate sampling techniques applied .A sampling error is the difference between a sample statistic and its corresponding parameter.

We can make probabilistic statements about this sampling error only if we have a probability sample.

Non-sampling error: Errors in observation, interview or measurement error, errors due to non-response and errors in data processing: editing, coding, etc.

The non-sampling error is likely to increase with increase in sample size. For instance, a census survey may have non-sampling errors in large amount collected in the course of an investigation. It should be selected using some predefined sampling technique in such a way that they represent the population very well.

1.2 Reasons for sampling

Sample survey saves money

Sample Survey saves time

Sample survey provides higher level of accuracy

Sample survey could be the only option for the study in some specialized area

Experimentation could be destructive in nature like testing industrial products such as testing the average duration of burning of bulbs, and testing the quality of wine, beer, etc.

In this case sampling is the only feasible means of study.

1.3 Sampling Techniques

The commonly used sampling techniques may be broadly classified as

Non Probability and Probability Sampling.

A. Random Sampling or probability sampling.

Probability sampling techniques is a method of sampling in which all elements in the population have a pre-assigned probability to be included in to the sample.

a/ Simple random sampling

b/ Stratified random sampling

c/ Cluster sampling

d/ Systematic sampling

a) Simple Random Sampling

A simple random sample from a population is a sample chosen randomly, so that each possible sample has the same probability of being chosen.

In small populations such sampling is typically done "without replacement", i.e., one deliberately avoids choosing any member of the population more than once.

Simple random sampling is a method of selecting n units out of a finite population of size N by giving equal probability to all units, or a sampling procedure in which all possible combinations of n units that may be formed from the finite population of size N units have the same probability of selection.

i) Lottery method

Example 1.4: If we want to take a sample of 25 persons out of a population of 150, the procedure is to write the names of all the 150 persons on separate slips of papers, fold these slips, mix them thoroughly and then make a blindfold selection of 25 slips without replacement.

b/ Stratified random sampling

- In stratified sampling, the population of N units is sub-divided into k sub-populations, called strata, so that the units in each stratum are as homogeneous as possible and the means of the different strata are as different as possible.
- $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$, where  represent the population size in the $t_i = \frac{N_i}{N}$ strata.
- Then a sample is drawn from each stratum independently, $H_A: \mu_2 > \mu_1$. The procedure of taking samples in this way is called stratified sampling.
- If the sample is taken randomly from each stratum, the procedure is known as stratified random sampling.

Remarks:

In stratified random sampling, the following two points are equally important to ensure accuracy:

- a) proper stratification of the population into various strata, and
- b) A suitable sample size from each stratum.

For example a population can be stratified based on the following variables:

- Sex (male, female)
- Age (under 18, 18 to 28, 29 to 39)
- Occupation (professional, other)
- Geographical classifications; Administrative regions, etc.

c/ Cluster Sampling:

The population is divided into non-overlapping groups called clusters.

A simple random sample of groups or cluster of elements is chosen and all the sampling units in the selected clusters will be surveyed in the case of single stage cluster sampling.

Clusters are formed in a way that elements within a cluster are heterogeneous, i.e. observations in each cluster should be more or less dissimilar. Cluster sampling is useful when it is difficult or costly to generate a simple random sample. For example, to estimate the average annual household income in a large city we use cluster sampling, because to use simple random sampling we need a complete list of households in the city as sampling frame.

d/ Systematic Sampling:

Systematic sampling is the selection of every k^{th} element from a sampling frame, where k , the sampling interval and $k = \text{population size} / \text{sample size} = N/n$.

Using this procedure each element in the population has a known and equal probability of selection. This makes systematic sampling functionally similar to simple random sampling. It is however, much more efficient and much less expensive to do. Like simple random sampling a complete list of all elements within the population (sampling frame) is required.

$k = \text{population size} / \text{sample size} = N/n$.

The procedure starts in determining the first element to be included in the sample, select a unit i randomly from the first group, $i \leq k$ as the first element. The second unit will be $(i+k)^{\text{th}}$ element from the frame. Totality we have a sample of size n from the population of size N , i^{th} , $(i+k)^{\text{th}}$, $(i+2k)^{\text{th}}$, ... $(i+(n-1)k)^{\text{th}}$ element of the population are taken as a sample.

Example 1.6

Suppose that $N = 20$ and we want to select a sample of size 4, so that $k = N/n = 20/4 = 5$.

The first element in the sample is selected from the first 5 units randomly, say 3^{rd} , which is the random start. Then, every 5^{th} unit is selected, and the sample contains the 3^{rd} , 8^{th} , 13^{th} and 18^{th} units of the population.

B. Non-Random Sampling or non-probability sampling.

It is a sampling technique in which the choice of individuals for a sample depends on the basis of convenience, personal choice or interest.

Types of non-random sampling are:

1. Judgment sampling.
2. Convenience sampling
3. Quota Sampling.

1. Judgment Sampling

In this case, the person taking the sample has direct or indirect control over which items are selected for the sample. This method is mainly used for opinion surveys but is not recommended for general use, as it bias of the sampler.

2. Convenience Sampling

In this method, the decision maker selects a sample from the population in a manner that is relatively easy and convenient.

3. Quota Sampling

In a quota sample, quotas are set up according to some specified characteristics such as income groups, age groups, political or religious groups, etc. Within the quota, the selection of sampling units depends up on personal judgment.

1.4 Sampling Distribution of the sample mean

Consider all possible samples of size n that can be drawn from a given population (either with or without replacement). For each sample, we can compute a statistic (such as the mean & the standard deviation) that will vary from sample to sample. In this manner we obtain a distribution of the statistic that is called its sampling distribution.

Steps for the construction of Sampling Distribution of the mean

1. From a finite population of size N , randomly draw all possible samples of size n . There are N^n possible samples if sampling is with replacement and there are $\alpha=0.05$ possible samples if sampling is without replacement.
2. Calculate the mean for each sample.
3. Summarize the mean obtained in step 2 in terms of frequency distribution

Example 1.7: Suppose we have a population of size 5, consisting of the age of five children 3, 5, 7, 9, and 11. Population mean is 7 and population variance is 8. (Consider sampling without replacement).

Take samples of size 2 and construct sampling distribution of the sample mean.

Solution:

Step 1: $N=5$, $n=2$ we have ${}^5P_2 = 10$, possible samples.

(3,5), (3,7), (3,9), (3,11), (5,7), (5,9), (5,11), (7,9), (7,11) and (9, 11)

Step 2: Calculate the sample mean for each sample:

Means = 4, 5, 6, 7, 6, 7, 8, 8, 9, 10 respectively.

Step 3: Summarize the mean obtained in step 2 in terms of frequency distribution.

$\bar{x}$	4	5	6	7	8	9	10	Total
f_i	1	1	2	2	2	1	1	10
f_i	4	5	12	14	16	9	10	70
$f(-7)^2$	9	4	2	0	2	4	9	30

a) Mean of sample means, $E(\bar{x}) = \frac{\sum f_i}{\sum f_i} = 70/10 = 7$

b) Variance of sample means, $\text{var}(\bar{x}) = \frac{\sum (-E(\bar{x}))^2}{k} = 30/10 = 3$

$$n_1 = n_2 = 3$$

Example 1.8: Three students have taken a class test which is marked out of 10. We want to estimate the mean mark using the sample mean as the estimate of the population mean. We take a sample of size 2 in two cases and suppose the marks of the three students are 1, 2 and 6.

The population mean μ is $(1+2+6)/3 = 3$

The population variance $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} = 14/3$.

i) Sampling without replacement

In this type of sampling an observation is included in the sample only once and is selected randomly without any preference or conscious effort.

If sampling is without replacement we can take ${}^3C_2 = 3$ possible samples; the possibilities are given below.

Possible sample	(1,2)	(1,6)	(2,6)
Sample mean	1.5	3.5	4

The sample mean is a random variable, and we see that it can take three possible values. We can now write down its probability distribution as follows

n_1	1.5	3.5	4	Total
$P(n_2 = p_1)$	1/3	1/3	1/3	1
$(-3)^2$	2.25	0.25	1	3.5

i) Mean of sample means $E\left(\frac{x_1 + x_2}{n_1 + n_2}\right) = \sum \bar{X}_i P(\bar{X}_i) = 1.5(1/3) + 3.5(1/3) + 4(1/3) = 3 = \text{population mean}$
i.e., Mean of sample means $E(\bar{p}) = \text{population mean}$

ii) Variance of sample means, $\text{var}(\bar{p}) = \frac{\sum (-E(\bar{p}))^2}{k} = 3.5/3 = 1.17$

where k is number of sample mean.

In which if sampling without replacement, $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = 14/12 = 1.17$.

ii) Sampling with replacement

In this type of sampling an observation has a chance to be selected at each draw.

Suppose that we take the sample with replacement, there are $3^2 = 9$ possible samples.

Sample	(1,1)	(1,2)	(1,6)	(2,1)	(2,2)	(2,6)	(6,1)	(6,2)	(6,6)
Sample mean	1	1.5	3.5	1.5	2	4	3.5	4	6

The sample mean is a random variable & its probability distribution is:

$\frac{x_1 + x_2}{n_1 + n_2}$	1	1.5	2	3.5	4	6	Total
$P\left(\frac{\hat{p}_1 - \hat{p}_2}{s_{\hat{p}_1 - \hat{p}_2}} = \frac{\hat{p}_1 + \hat{p}_2}{n_1 + n_2}\right)$	1/9	2/9	1/9	2/9	2/9	1/9	1
$\sum_{i=1}^k (-E(\bar{p}))^2 P(\bar{p})$	1/9	1/3	2/9	7/9	8/9	6/9	3
$f_i(-3)^2$	4	4.5	1	0.50	2	9	21

i) Mean of sample means $E(\bar{p}) = \sum \bar{X}_i p_i$
 $= 1(1/9) + 1.5(2/9) + 2(1/9) + 3.5(2/9) + 4(2/9) + 6(1/9) = 3$

Mean of sample means, $E(\bar{p}) = \text{population mean}$.

ii) Variance of sample means $\text{var}(\bar{p}) = \frac{\sum (\bar{X}_i - E(\bar{p}))^2}{k} = 21/9 = 2.33$

Where k is number of sample means

$H_0: p_{1978} - p_{1979} = 0$
 $= \frac{H_A: p_{1978} - p_{1979} \neq 0}{14/6} = 14/6 = 2.33$

In which if sampling with replacement, $H_A: p_{1978} - p_{1979} \neq 0$
 $= \frac{14/6}{0.01} = 14/6 = 2.33$

In each case the expected value of the sample mean equals the population mean. This explains why the sample mean is a good estimate of the population mean. If we use the sample mean as an estimate of the population mean we will sometimes overestimate it, and sometimes underestimate it, but “on average” we will be accurate.

The example above illustrates an important result.

Remark:

1. Mean of sample means $= E(\bar{p}) = \frac{\sum f_i}{\sum f_i} = \sum \bar{X}_i p_i = \text{population mean}$

2. Variance of sample means, $\sqrt{(0.0308)(0.9692) \left(\frac{1}{15,000} + \frac{1}{10,000} \right)}$ (if sampling is with replacement)

3. Variance of sample means, H_0 , (if sampling is without replacement)

The quantity $H_0: \sigma^2 = \sigma_0^2$ is finite population correction (fpc), and if $n/N < 0.05$, fpc is ignored.

Note: the square root the variance of sample means is known as **standard error**.

1.5 Sampling Distribution of the sample Proportion

In situations where it is not possible to measure the characteristic under study, but is possible to classify the whole population in various categories with respect to the attributes they possess, consideration is usually given to estimating the population elements that belong to a defined category of class.

N.B: $\text{var}(\hat{p}) = \frac{xy}{n} = \frac{y^2}{n}$

Where sampling fraction, $f = n/N$

$q = 1-p$

The sampling fraction can be ignored, when N is large relative to sample size n , $n/N < 0.05$.

$\text{var}(\hat{p}) = \frac{28.375}{58.875} = 0.482$ and the standard error of p is $\sqrt{0.482}$.

Example 1.12

- In a simple random sample of size 100, from a population of size 500, there are 37 employed persons in the sample.

a) Estimate proportion of employed persons in the population.

b) Calculate the standard error of p .

Solution:

a) Population proportion P is estimated by $p = a/n = 37/100 = 0.37$ of the population is employed.

b) Standard error of p is $\sqrt{pq} = \sqrt{0.37 \times 0.63} = 0.482$.